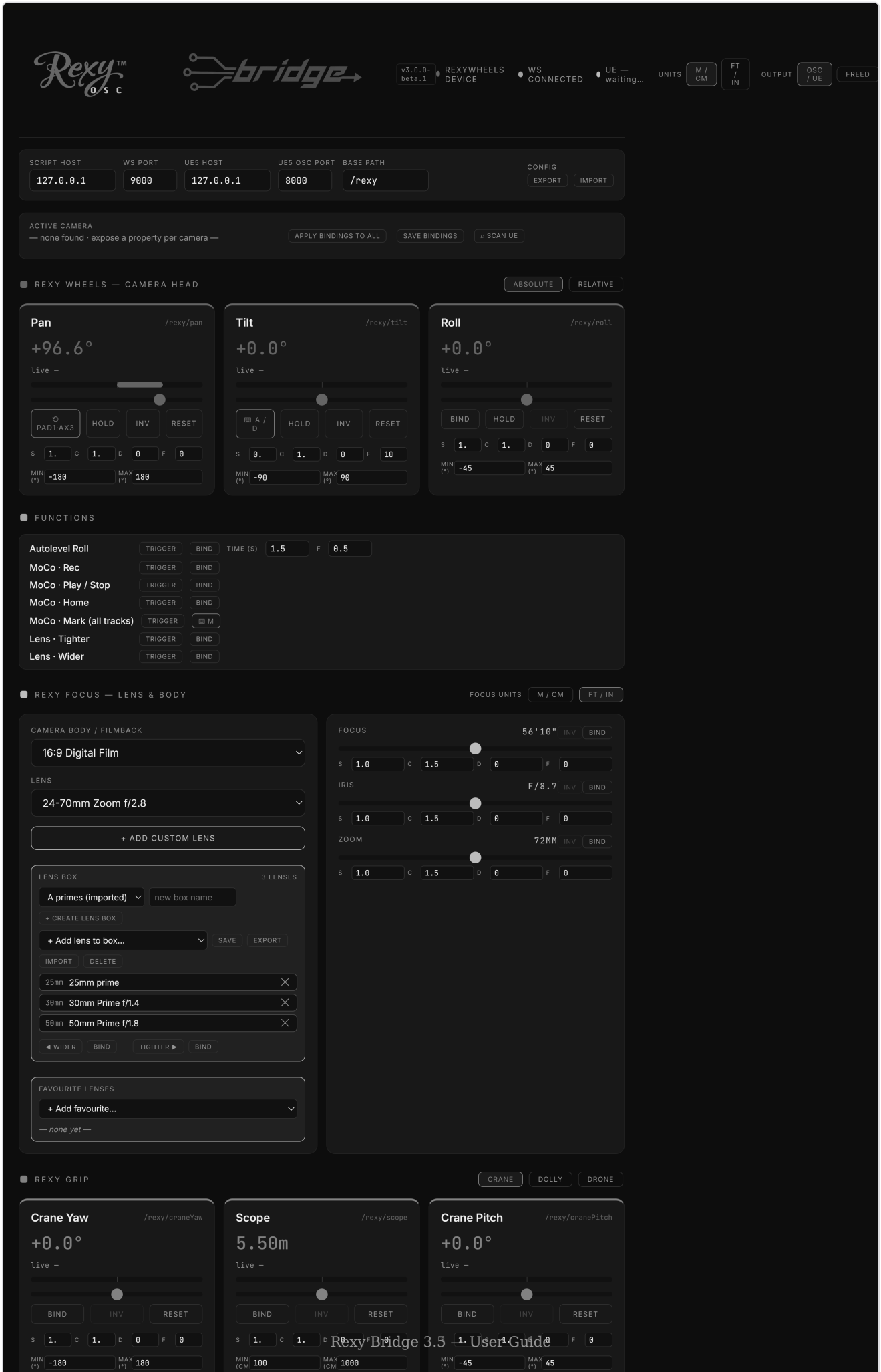
Rexy Bridge 3.5 — User Guide

A panel-by-panel tour of every control, in the order they appear in the app. Setup and UE connection are covered in **GETTING_STARTED.md**; this guide assumes you're connected and driving a camera.


Rexy OSC
bridge
v3.0.0-beta.1
REXYWHEELS DEVICE
WS CONNECTED
UE — waiting...
UNITS
M / CM
FT / IN
OUTPUT
OSC / UE
FREED
SCRIPT HOST
127.0.0.1
WS PORT
9000
UE5 HOST
127.0.0.1
UE5 OSC PORT
8000
BASE PATH
/rexy
CONFIG
EXPORT
IMPORT
ACTIVE CAMERA
— none found · expose a property per camera —
APPLY BINDINGS TO ALL
SAVE BINDINGS
SCAN UE
REXY WHEELS — CAMERA HEAD
ABSOLUTE
RELATIVE
Pan
/rexy/pan
+96.6°
live —
PAD1·AX3
HOLD
INV
RESET
S 1. C 1. D 0 F 0
MIN (°) -180
MAX (°) 180
Tilt
/rexy/tilt
+0.0°
live —
A / D
HOLD
INV
RESET
S 0. C 1. D 0 F 10
MIN (°) -90
MAX (°) 90
Roll
/rexy/roll
+0.0°
live —
BIND
HOLD
INV
RESET
S 1. C 1. D 0 F 0
MIN (°) -45
MAX (°) 45
FUNCTIONS
Autolevel Roll
TRIGGER
BIND
TIME (S) 1.5
F 0.5
MoCo · Rec
MoCo · Play / Stop
MoCo · Home
MoCo · Mark (all tracks)
M
Lens · Tighter
Lens · Wider
REXY FOCUS — LENS & BODY
FOCUS UNITS
M / CM
FT / IN
CAMERA BODY / FILMBACK
16:9 Digital Film
LENS
24-70mm Zoom f/2.8
+ ADD CUSTOM LENS
LENS BOX
3 LENSES
A primes (imported)
new box name
+ CREATE LENS BOX
+ Add lens to box...
SAVE
EXPORT
IMPORT
DELETE
25mm 25mm prime
30mm 30mm Prime f/1.4
50mm 50mm Prime f/1.8
◀ WIDER
BIND
TIGHTER ▶
BIND
FAVOURITE LENSES
+ Add favourite...
— none yet —
FOCUS
56'10"
INV
BIND
S 1.0 C 1.5 D 0 F 0
IRIS
F/8.7
INV
BIND
S 1.0 C 1.5 D 0 F 0
ZOOM
72MM
INV
BIND
S 1.0 C 1.5 D 0 F 0
REXY GRIP
CRANE
DOLLY
DRONE
Crane Yaw
/rexy/craneYaw
+0.0°
live —
BIND
INV
RESET
S 1. C 1. D 0 F 0
MIN (°) -180
MAX (°) 180
Scope
/rexy/scope
5.50m
live —
BIND
INV
RESET
S 1. C 1. D 0 F 0
MIN (CM) 100
MAX (CM) 1000
Crane Pitch
/rexy/cranePitch
+0.0°
live —
BIND
INV
RESET
S 1. C 1. D 0 F 0
MIN (°) -45
MAX (°) 45

Core concepts (read this first)

Binding. Almost every control has a **Bind** button. Click it, then actuate the input you want — a wheel, a stick, a keyboard key, a MIDI knob, or an OSC/FreeD channel. For continuous controls, push it the way you want to be "positive" first; for keys/buttons it captures a two-stage *increase* then *decrease*. Right-click a Bind button to clear it.

Per-camera bindings. Every camera you select keeps its **own** set of bindings. Switch cameras and your inputs re-map to that camera's saved layout.

S / C / D / F tuning. Wherever you see these four little boxes, they shape how an input drives the value:

- **S — Sensitivity / Speed.** Multiplies how fast the control moves the value. Works for analogue *and* keyboard/button inputs (higher = faster).
- **C — Curve.** Response gamma. `1.0` = linear; higher = "fluid-head" feel (gentle at first, faster as you push); lower = punchy.
- **D — Deadband.** Ignores tiny inputs near centre — kills stick/wheel drift. (Not relevant to keyboard/button inputs.)
- **F — Feather.** Smooths the start and stop so motion eases in and out instead of snapping.

Inv / Hold / Reset. Inv flips a binding's direction. **Hold** (on wheels) freezes that axis so its binding is ignored. **Reset** returns the control to centre/default.

1. Top bar



- **Version badge** (`v3.5.0-beta`) — the running build; it now **increments every dev build**, so it tells you at a glance exactly which build you're on. If it's missing or wrong after an update, press **Ctrl+Shift+R**.
- **⟲ Reset** — clears all saved settings and bindings on this machine and reloads from defaults. Use it if a new build seems to hang on stale data; your exported `.json` config files are **not** touched.
- **HARDWARE** dot — your gamepad/wheels. Wakes when you click the window and press a physical button.
- **WS** dot — the internal app↔bridge link.
- **UE RC** dot — the Unreal Remote Control connection.
- **Units** — global metric (m/cm) vs imperial (ft/in) for distances and positions.
- **Dark / Light** — switches the whole interface between the default dark theme and a pale blue-grey light theme. Your choice is remembered, and every accent colour stays the same in both.

- **Check for updates** — manually checks the release server for a newer build; if one exists it downloads in the background and offers **Restart to update**. Nothing is ever checked automatically on launch.
- **Font** — S / M / L text size for the whole interface; your choice is remembered.

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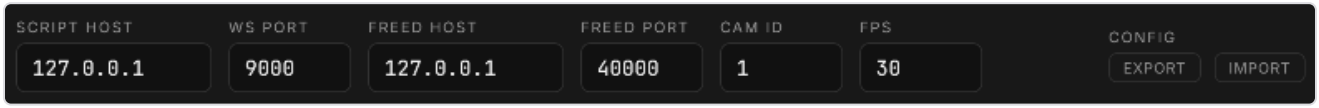
2. Output mode & network

The **osc / ue** vs **freed** toggle chooses what Rexy Bridge drives, and the network fields beneath it change to match.

- **osc / ue** (default) — drives Unreal directly via Remote Control. The fields show the UE/OSC target host and port.



- **freed** — broadcasts a **FreeD** camera-tracking stream on the network so any FreeD-compatible app can consume it (no UE writes). The fields switch to the FreeD stream host/port.

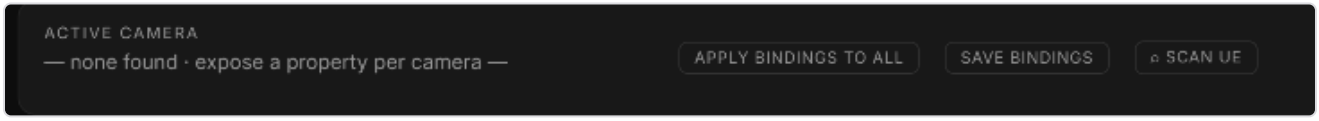


Your connection settings, along with every binding and tuning value, save and reload from the Config controls — export a `.json` to move a whole setup between machines.

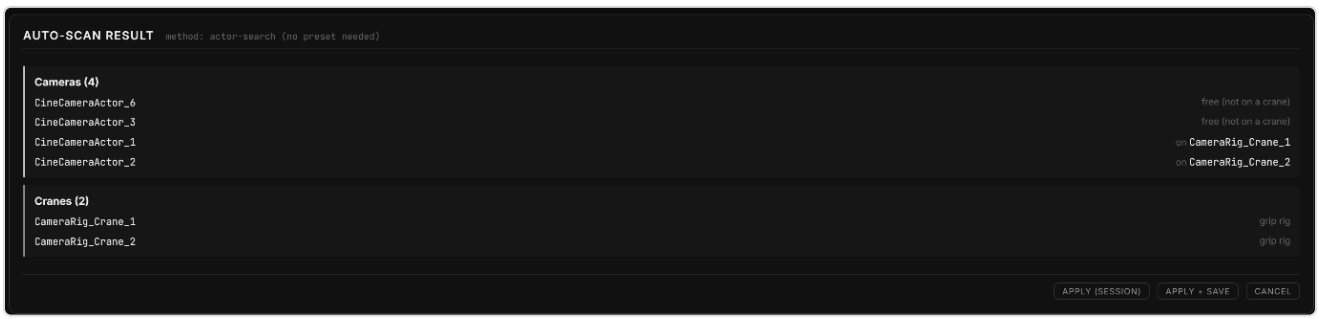
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3. Cameras — Scan & select

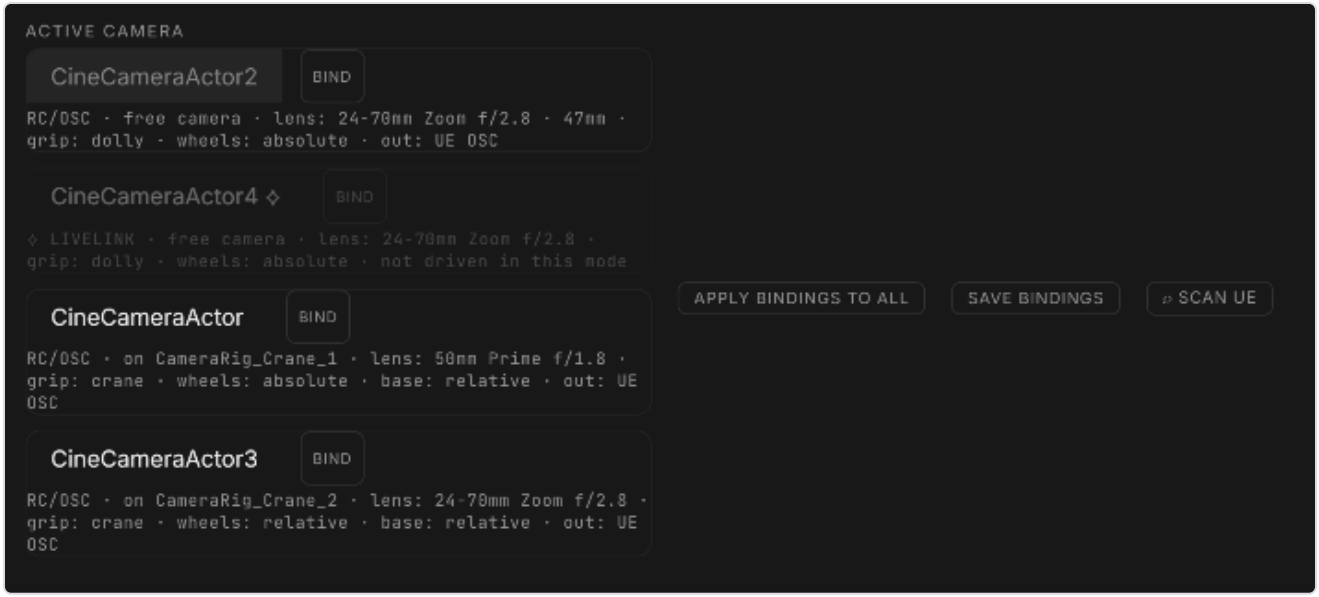
Before scanning, the camera row is empty and waiting.



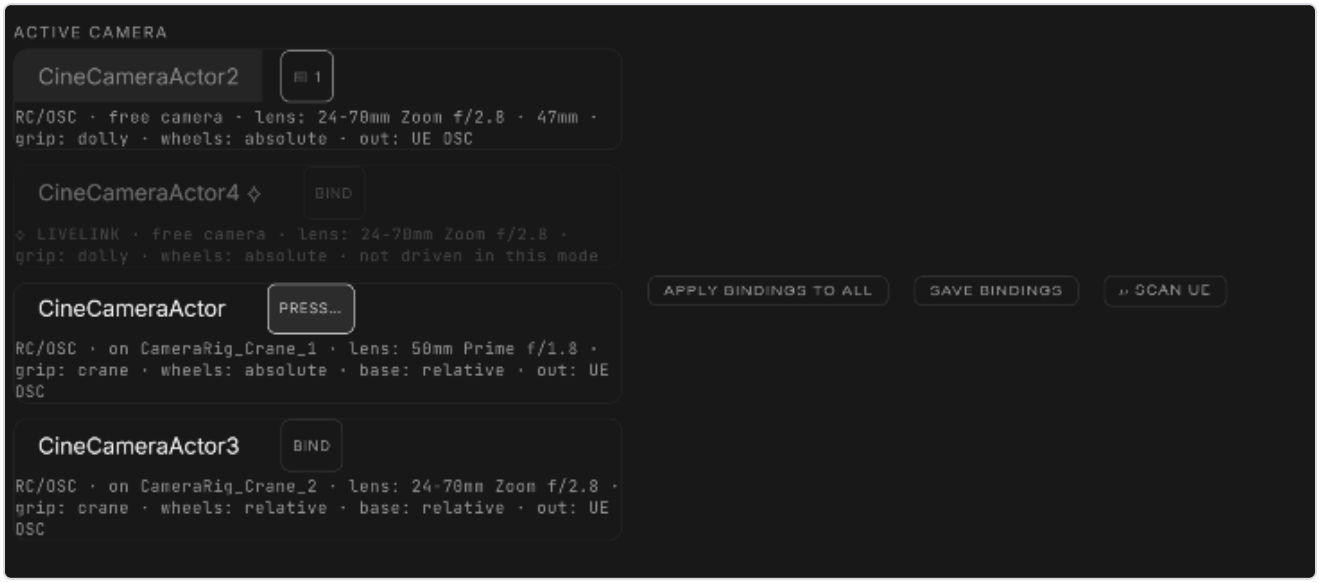
Click ⌕ **Scan UE** to auto-discover every Cine Camera Actor and CameraRig_Crane in your open level — no RexyControl preset, no `mappings.json` editing. Actors are matched by **class**, so it finds them whatever you've named them in the Outliner. Review the result, then **Apply** for this session or **Apply + Save** to persist it.



Discovered cameras appear as buttons. Click one to make it the **active** camera; its bindings and profile load.

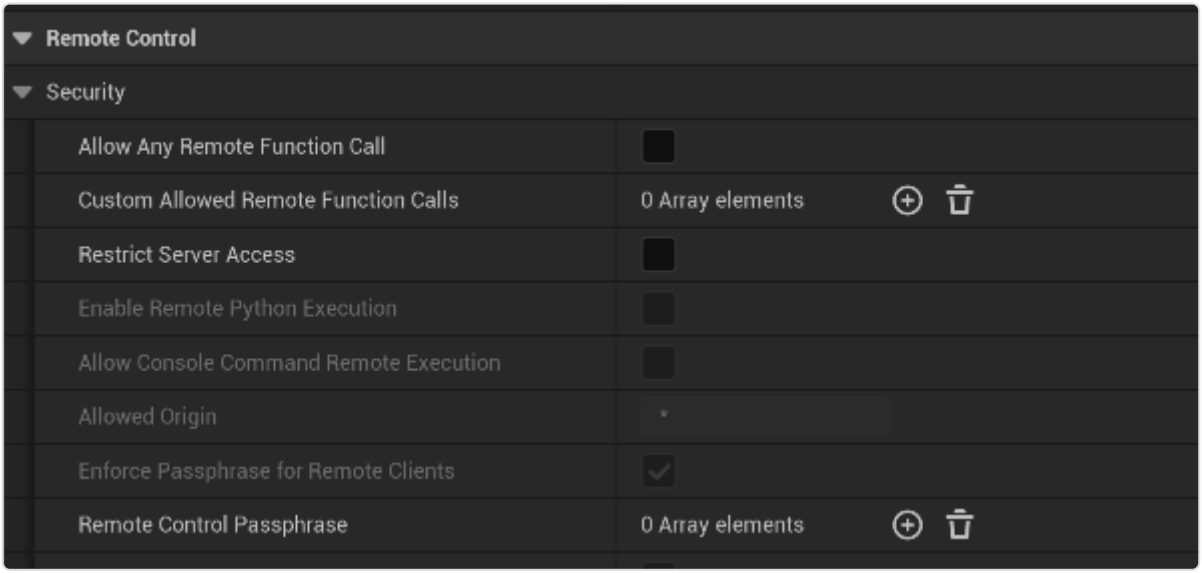


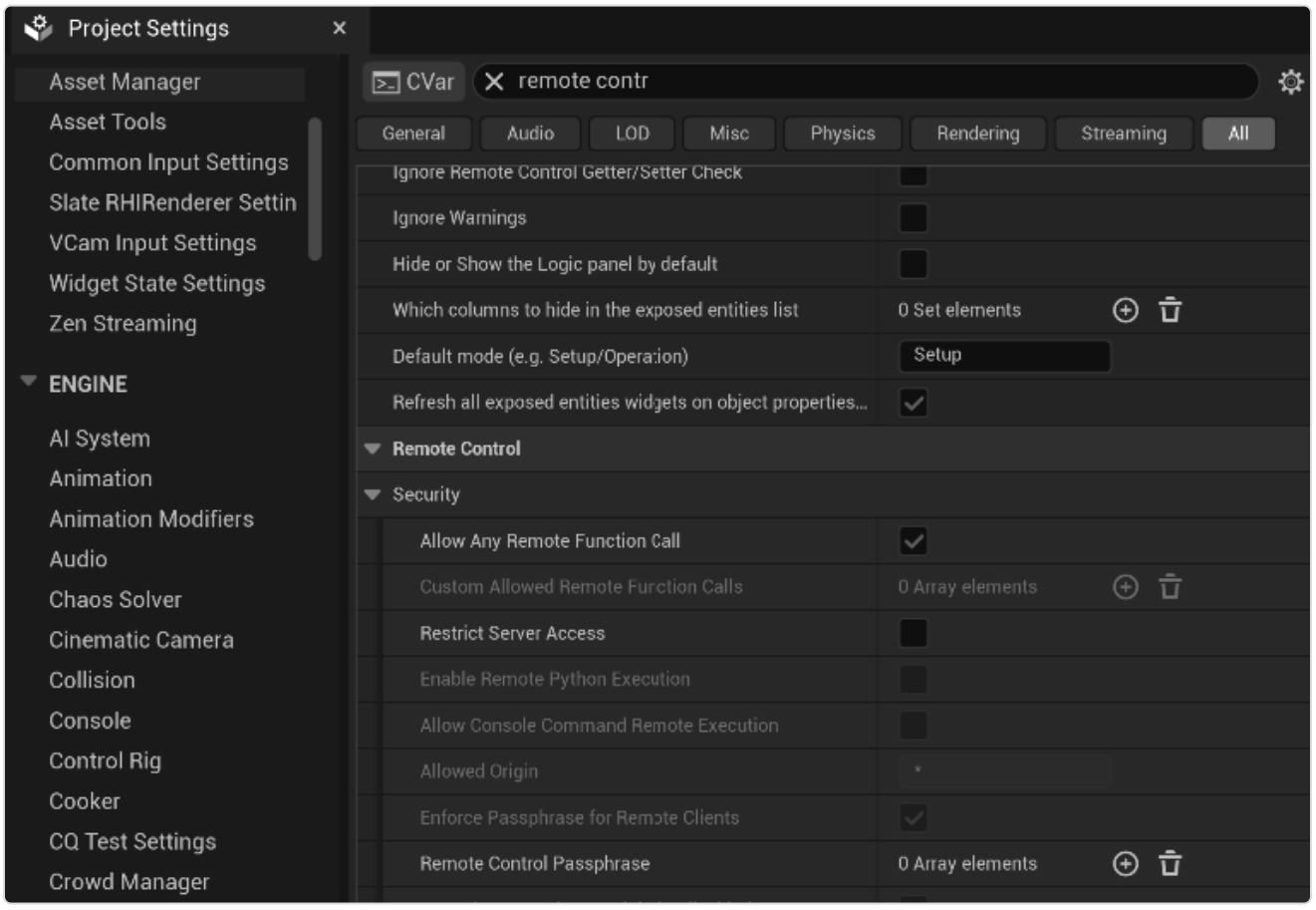
Each camera button can itself be **bound** to a key or gamepad button for hands-free switching mid-shot.



On UE 5.8 — allow remote function calls

UE 5.8 blocks remote function calls by default, so **Scan UE finds nothing** until you turn this on (one time, per project). In UE: **Project Settings → search "Allow Any Remote Function Call" → tick it on**.

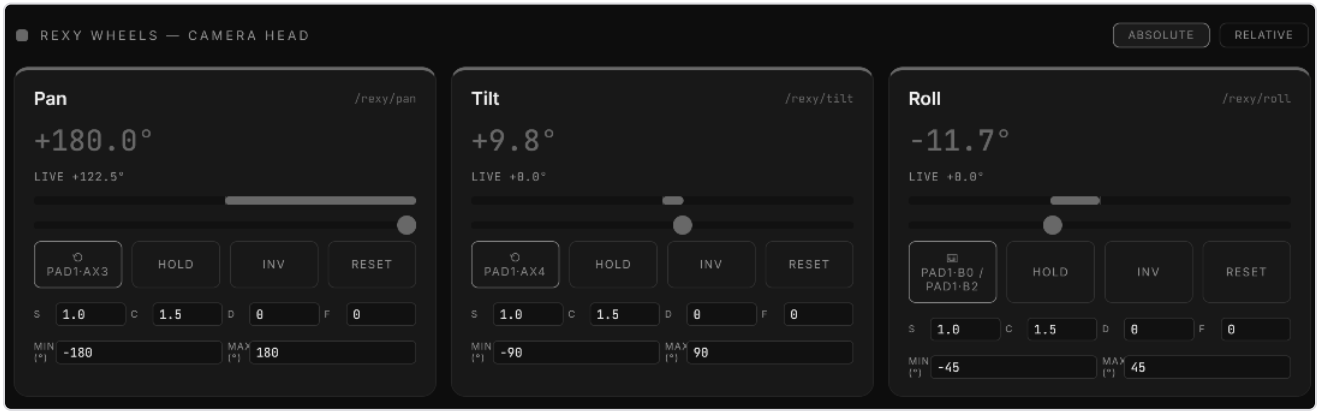




4. Rexy Wheels — camera head

The three head axes: **pan, tilt, roll**.

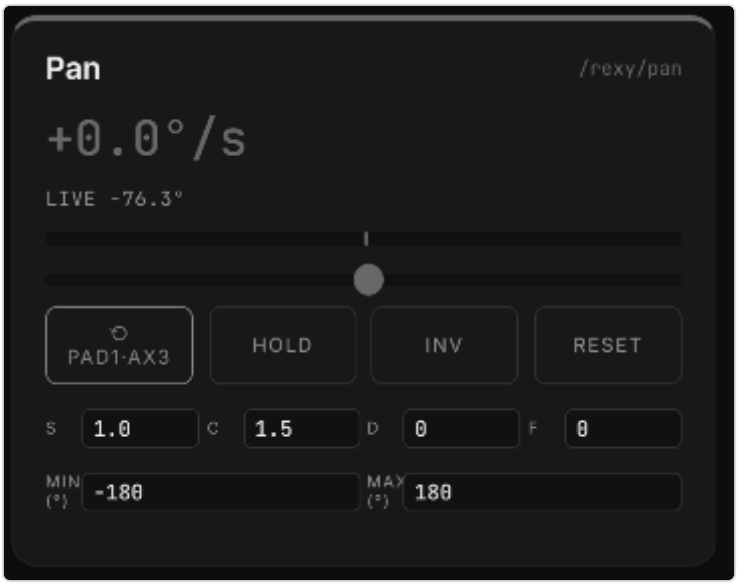
- **Absolute** — wheel position = camera angle.



- **Relative** — wheel deflection = rotation *speed*, centre = hold (endless-rotation, fluid-head style).

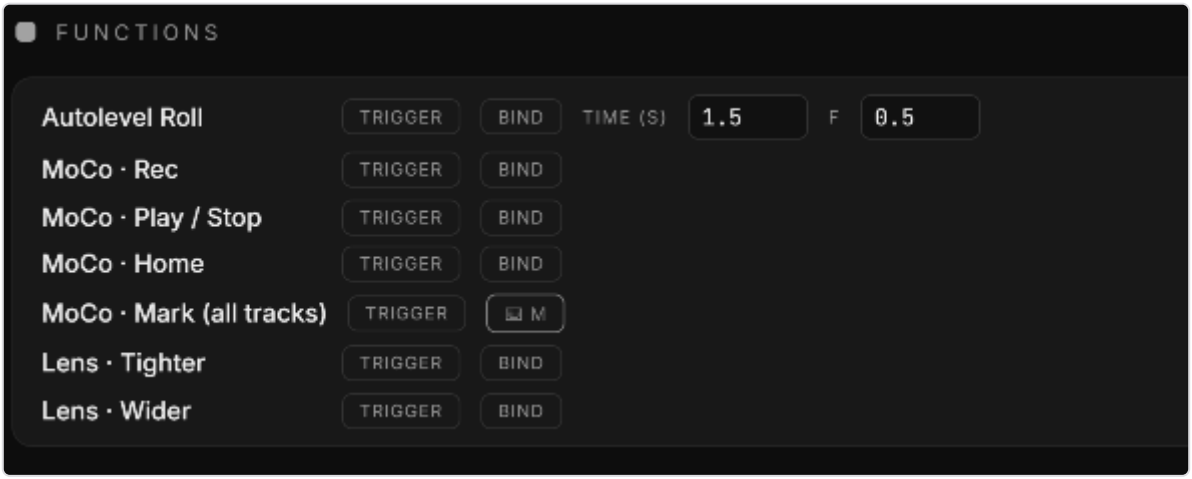


Each card has **Value** and **live —** readouts (commanded vs the camera's actual angle from UE), a **slider**, **Bind** / **Hold** / **Inv** / **Reset**, the **S** / **C** / **D** / **F** tuning row, and **Min** / **Max** to clamp the output range.



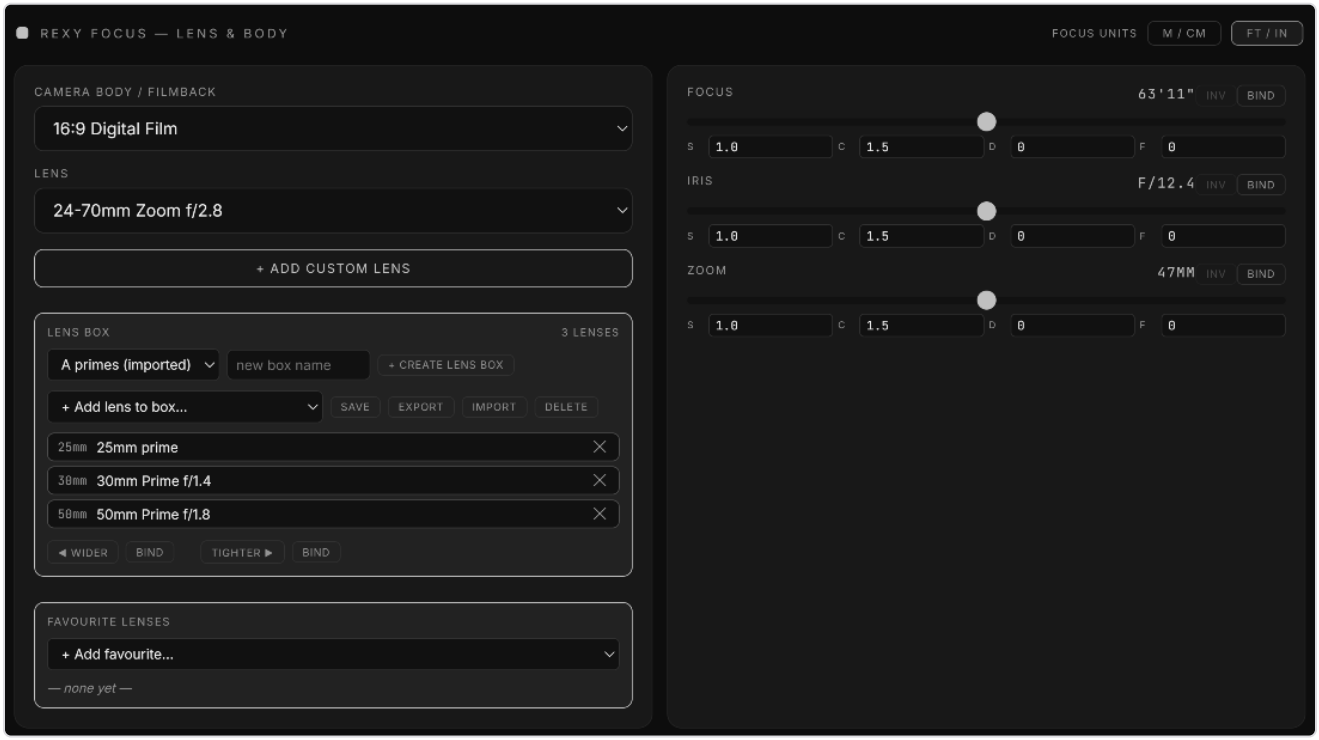
5. Functions

Bindable one-shot actions. Each card has a **Trigger** (fire it now) and a **Bind** (assign a key/button). Binding any copy of an action keeps every copy in sync.



- **Autolevel Roll** — smoothly returns roll to 0° over **Time** seconds, with an **F** easing.
- **MoCo · Rec** / **Play** / **Stop** / **Home** / **Mark** — transport actions, bindable so you never reach for the mouse mid-take. **Mark** drops a mark on **every bound track** at the playhead.
- **Lens · Tighter** / **Wider** — step to the next longer/shorter lens in the active **lens box**.

6. Rexy Focus — lens & body



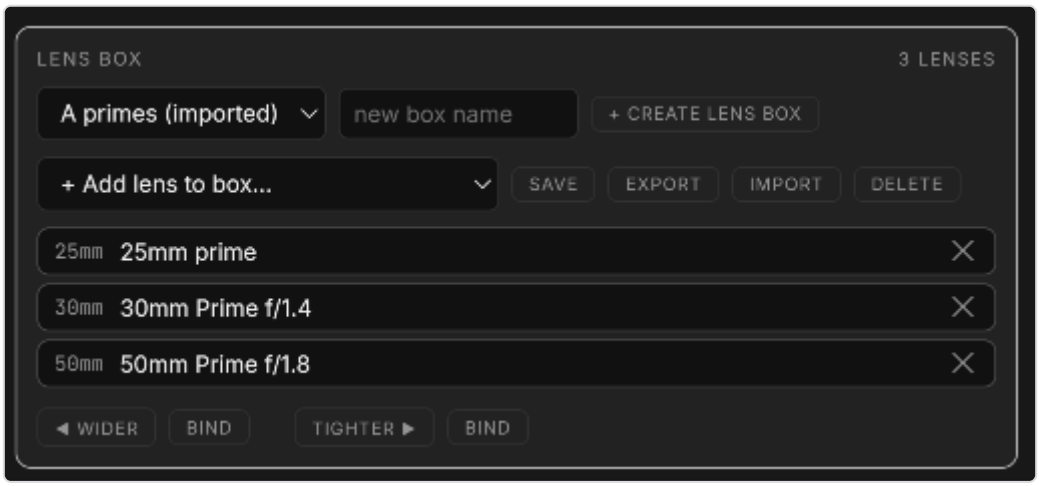
Lens & body

- **Camera body / filmback** — sensor/filmback preset.

- **Lens** — pick a lens; UE's LensSettings (focal, aperture, min-focus) update live.
- **+ Add custom lens** — define name, focal min/max, f-stop min/max, and **Min focus** (shown in cm or ft to match your Focus-units toggle). It's added to the list and applied.

Lens boxes

A **lens box** is a named kit — a list of lenses you cycle through on set.



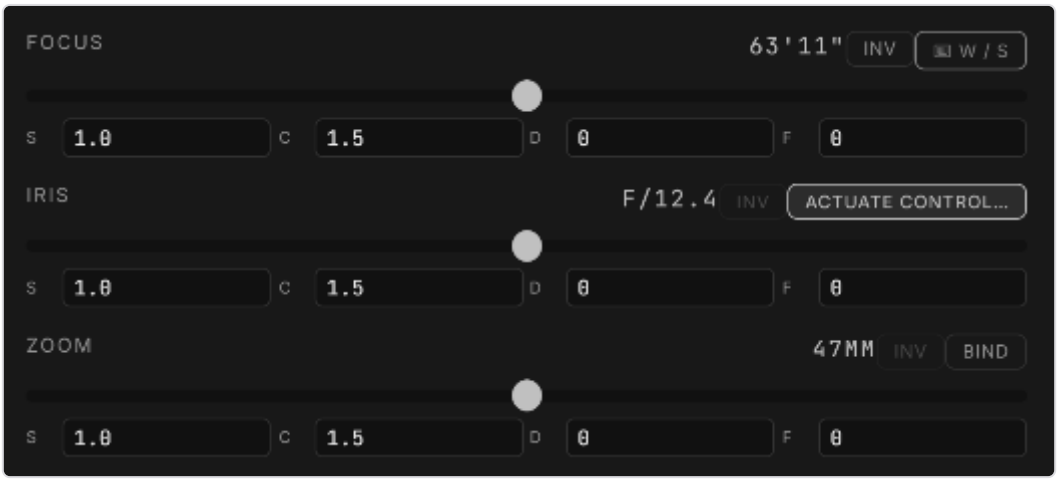
- **Box selector** + **Create lens box** — make and switch between kits.
- **Add lens to box** dropdown — add any existing lens, or ✎ **+ Custom lens…** to define a new one that drops straight into the box.
- **Lens chips** — the box's lenses, always sorted by focal length; the active lens is highlighted. × removes one.
- **◀ Wider / Tighter ▶** — step through the box by focal length (clamped at the ends). Each has a **Bind** for hardware control.
- **Save / Export / Import / Delete** — boxes export and import independently of the main config, so you can share a kit.

Favourite lenses



A global quick-jump list. Add any lens, then **Bind** a key/button to jump straight to it — independent of which box is loaded.

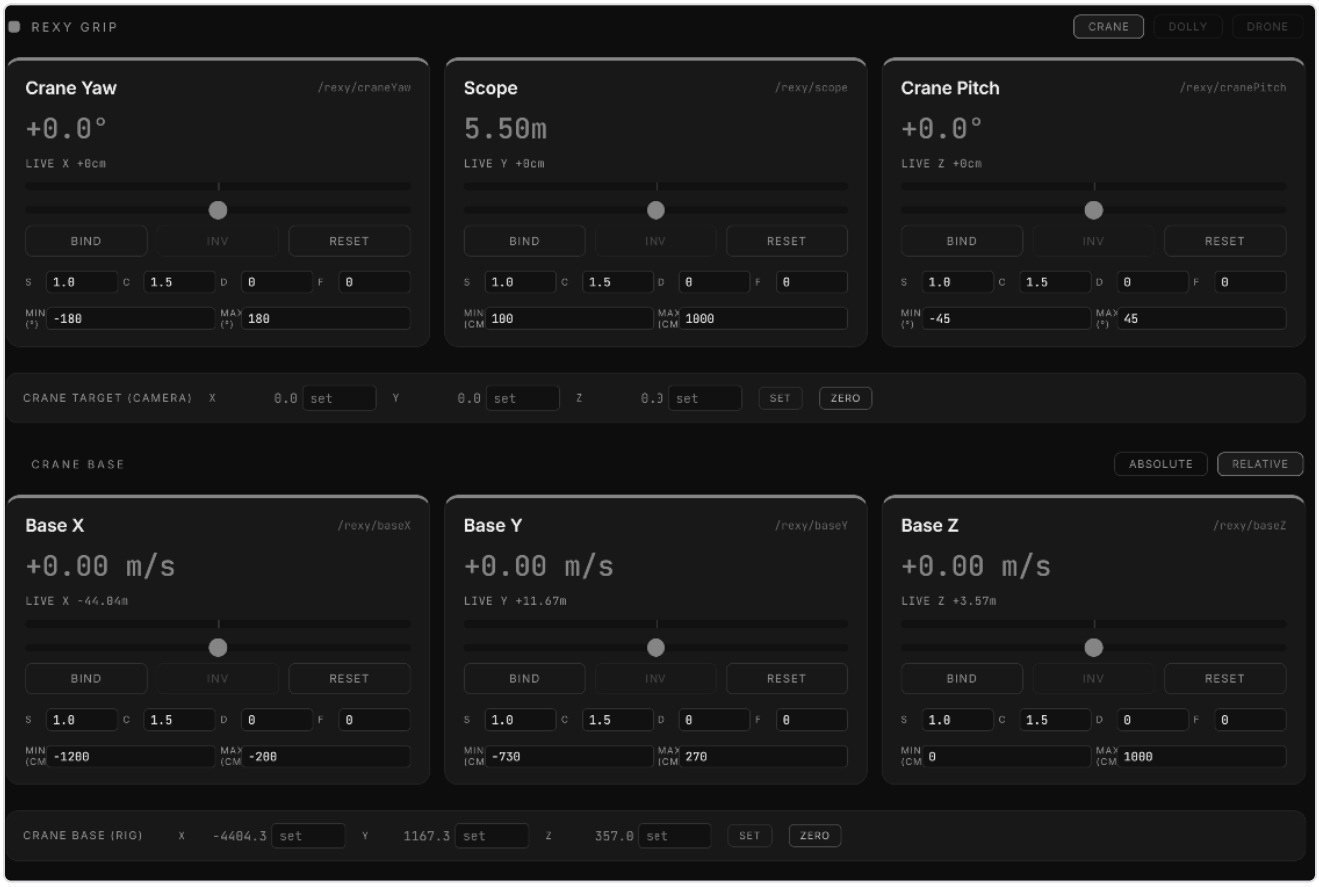
Focus / Iris / Zoom



Each has a slider, **Bind**, **Inv**, a live readout, and the full **S / C / D / F** tuning row — so a focus puller can bind a wheel or keys and tune the feel exactly like the head axes. Zoom locks automatically on a prime lens.

---

7. Grip — crane / dolly / drone



- **Crane** — the grip swings a jib arm around its base (yaw, pitch, scope/arm-length).
- **Dolly** — straight ground translation (X/Y/Z), no arm pivot.
- **Drone** — free flight through 3D space.

In **Drone** mode a **flight-frame** selector appears next to the mode buttons, with three options:

- **Cine** — flies on the world grid; you aim the camera independently with the wheels. Left/right, forward/back and up/down all follow the world axes, exactly like Dolly.
- **Cine/FPV** — left/right and forward/back follow **where the camera is panned**, but stay **parallel to the ground** regardless of tilt or roll; up/down is always a **true vertical** jib. So with the camera tilted 45° down, forward still travels flat in the pan direction, and a rolled camera still tracks level left/right. This is the natural "operator flying a level drone while framing freely" feel.
- **FPV** — everything is relative to the camera body: left/right, forward/back **and** up/down all follow the lens, so up/down rides the roll/tilt. True first-person-drone flight.

⟳ **Reset grid** (next to the frame buttons) re-snaps the Cine/FPV ground grid to the camera's current pan heading. In absolute-pan mode the grid follows the pan wheel live; use Reset if it ever drifts out of sync (e.g. after operating in continuous-wheel mode).

Hide Crane (in the Crane Base header, and as a bindable card in the Functions panel) vanishes the crane rig from both the render and the editor viewport, so the arm never appears in shot — press again to bring it back.

The grip cards (craneYaw, scope, cranePitch) use the same binding + S/C/D/F model as the wheels. When **Limits** is unticked, the per-axis Min/Max fields grey out and movement runs unclamped.

Real-world position limits — a world safe box, a box that follows the crane base, and a crane **height** cap — live in **Soft Stops & Camera Box** (section 9).

Crane Base

The base (baseX/Y/Z) drives the rig's pivot, with two modes:

- **Absolute** — slider position = base position between Min and Max.



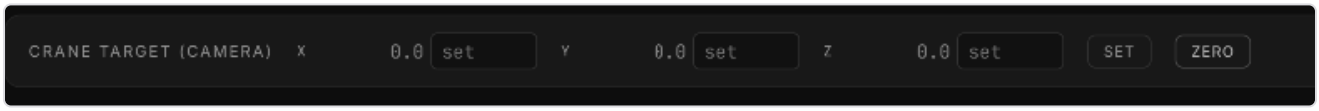
- **Relative** — slider deviation = movement speed (m/s); centre = stop.



8. Position panels

Live **X / Y / Z** readouts, with **Set** (send a manual position; blank axes keep their current value) and **Zero** (reset to 0,0,0).

- **Crane target** — where the camera/target sits.



- **Crane base** — where the rig base sits.



9. Soft Stops & Camera Box

Soft Stops and the Camera Box are the safety rails — they stop the camera *before* it reaches somewhere you don't want it, easing into the limit rather than hitting it. Both are enforced on the bridge, so they hold no matter what's driving the camera: wheels, grip, a recorded move, or a remote operator.

Soft Stops — per-parameter limits

Turn on the **master** switch, then enable a stop on any parameter and set its **Min**, **Max** and **Feather**:

- **Min / Max** — the range the value is held within, in the parameter's own units (degrees for the head and crane axes, cm for base and camera moves, mm / f-stop for the lens). The **live** button on each field captures the parameter's current value, so you can drive the camera to a limit and press it rather than typing a number.
- **Feather** — how gently the camera decelerates into the wall. `0` is a hard stop; higher eases it in over a wider zone, the same feel as the S/C/D/F feather.
- **Stop → Min / Stop → Max** — snap every enabled stop to where the camera is right now, in one click; both are bindable in the Functions panel.
- Stops are saved **per camera**, and grey out when they don't apply to the current mode (e.g. the base stops are crane-only).

Soft stops hold on **both** kinds of move. On **absolute** controls (pan / tilt / roll in Absolute wheel mode, focus / iris / zoom, crane yaw / scope / pitch, base in Absolute mode) the value is simply clamped. On **velocity** controls — continuous ("Relative") wheels and the crane base in Relative mode — the bridge tracks the accumulating angle or position and feathers the *speed* into the limit, so an endless-rotation pan decelerates and stops exactly at your degree limit and reverses straight back out.

Whenever an axis is up against a limit, its card flashes a **STOP** and its slider pins to the wall, so what you see always matches the real rig. The command stops dead at the limit instead of winding up past it, so the move reverses the instant you back off.

> The velocity walls read the rig's position once when you enter the mode. If you reposition the rig by hand in Unreal mid-take, re-enter the mode (flip Relative off/on, or re-pick the grip mode) to re-sync.

Camera Box — a real-world safe volume

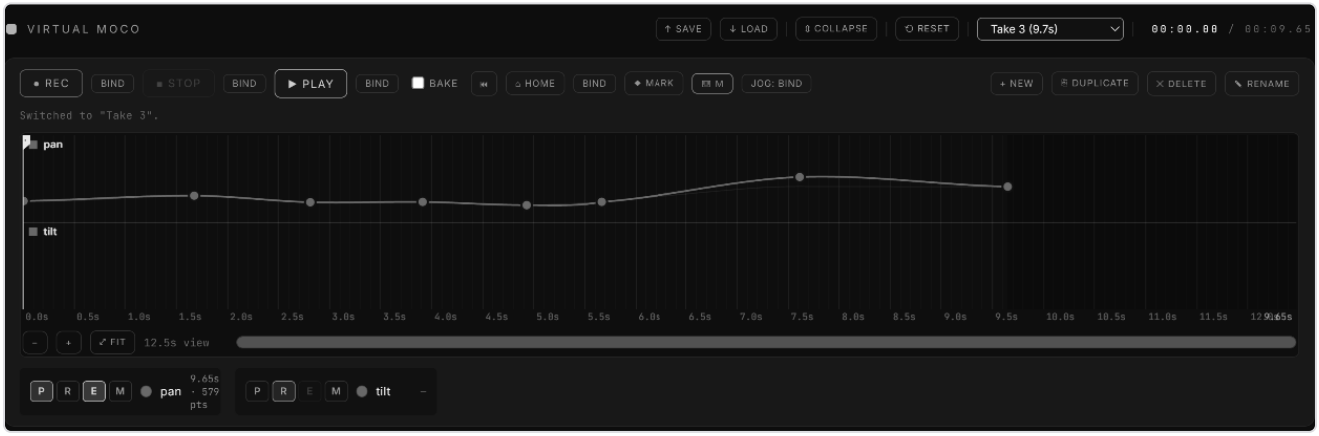
Below the per-parameter stops, the **Camera Box** limits the camera's **world position** — independent **X**, **Y** and **Z** limits in cm, each with its own on-switch and feather, enforced live in dolly and drone. Use it to keep the camera inside the set, above the floor, or under the lighting grid. Each axis has a **live** button that reads the camera's current world position.

- **Zero offset (X / Y / Z)** — the box's Min/Max are measured from this offset instead of world zero, so you can place the box anywhere in the world.
- **Follow crane base** — locks the offset to the crane base's live position, so the whole box **travels with the base**. That turns the world box into a *crane-local* envelope: the camera stays limited relative to the base wherever the base rolls.
- In **crane** mode the box's **Z axis caps the camera height**: the bridge works the height out from the arm length and pitch (`ArmLength × sin(pitch)`) and eases scope and pitch into your ceiling/floor, whichever you push. It auto-calibrates to your exact rig the moment you enter crane mode — you'll see a `Crane height calibrated` line in the log. (Horizontal reach from the base is best limited with the **Scope** grip soft stop.)

- In **FPV** drone flight the box becomes a **local** envelope — how far you've flown from the point you entered FPV, in the camera's own axes.

10. Virtual MoCo

Record, edit, and play back moves across every bound parameter.

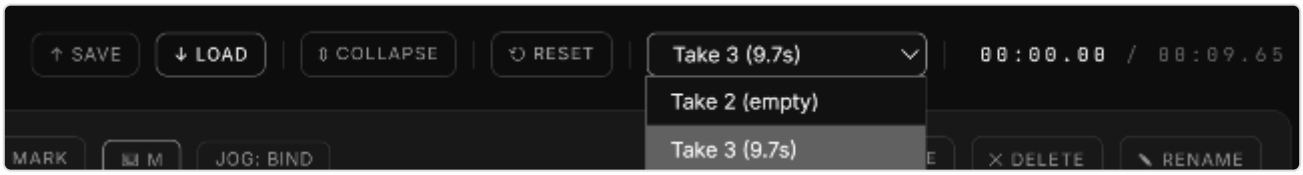


Transport



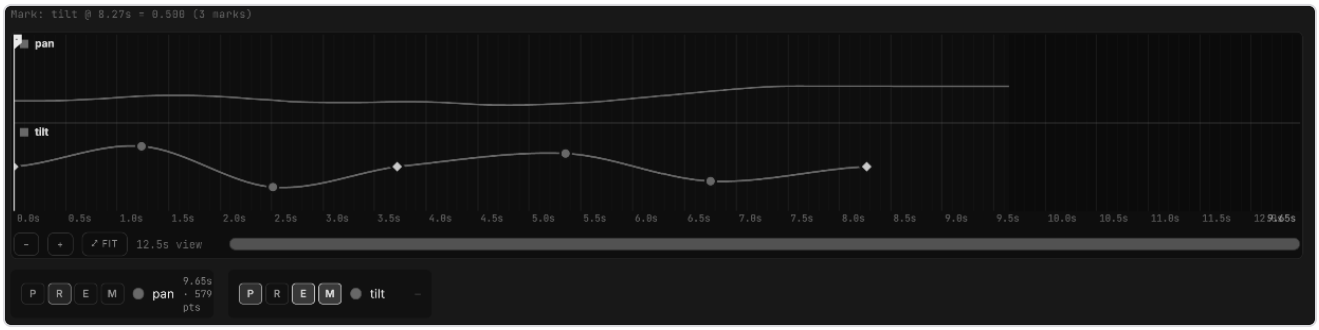
- **● REC / ■ STOP / ▶ PLAY** — each with a **Bind** button. STOP leaves the playhead where it is; **Home** (or Rewind) returns it to the start.
- **Bake** — a 3-2-1 countdown on REC/PLAY so you can sync with UE's Take Recorder.
- **⌂ HOME** — snaps play-armed tracks to their start values and the playhead to 0.
- **◆ MARK** — drops a mark on every bound track at the playhead.
- **Jog: Bind** — bind an analogue wheel/axis (or keys) to **shuttle** the playhead: push from centre to scrub forward/back, speed proportional to deflection, centre = hold.

Takes



Multi-take management. Save/Load moves as `.rxmove` files. **Expand** stacks each track in its own lane; **Reset** clears everything.

Timeline & tracks



Colour-coded per track. Scrub by dragging the playhead, zoom with the scroll wheel, and drag anchor dots to reshape curves. Each track (left of the timeline) has four arm toggles:

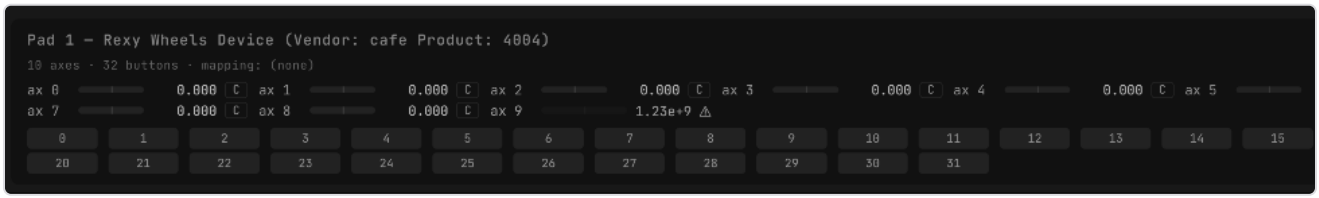
- **P — Play** — plays back its recorded/marked data.
- **R — Record** — captures live input on the next REC (recording over a track auto-erases its marks).
- **E — Edit** — drag anchor dots to reshape. Available once a track has recorded data **or 2+ marks** — entering E on a marked-but-unrecorded track lays down draggable anchors that smooth the marks into **S-curves**.
- **M — Mark** — drops a fixed point at the playhead; the curve always passes through marks. Placing a mark auto-adds a start mark at t=0 so the move has a defined beginning.

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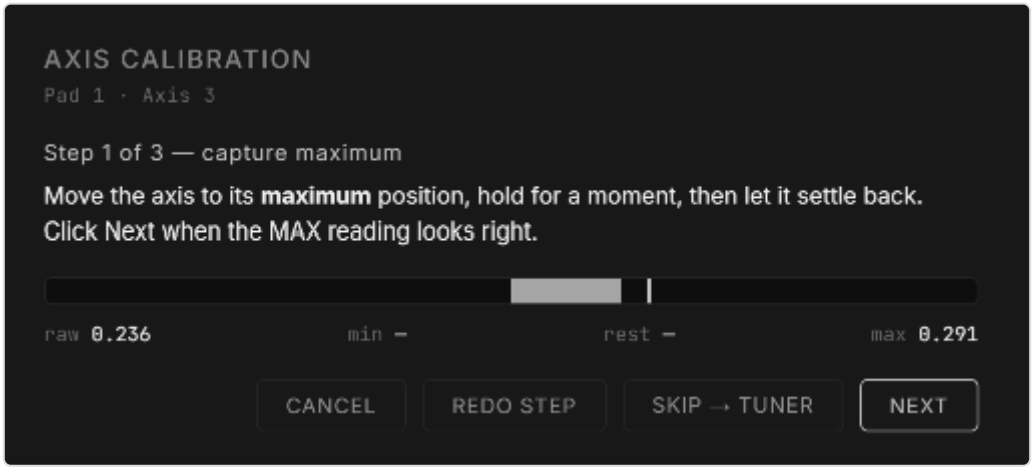
11. Gamepad Debug & calibration



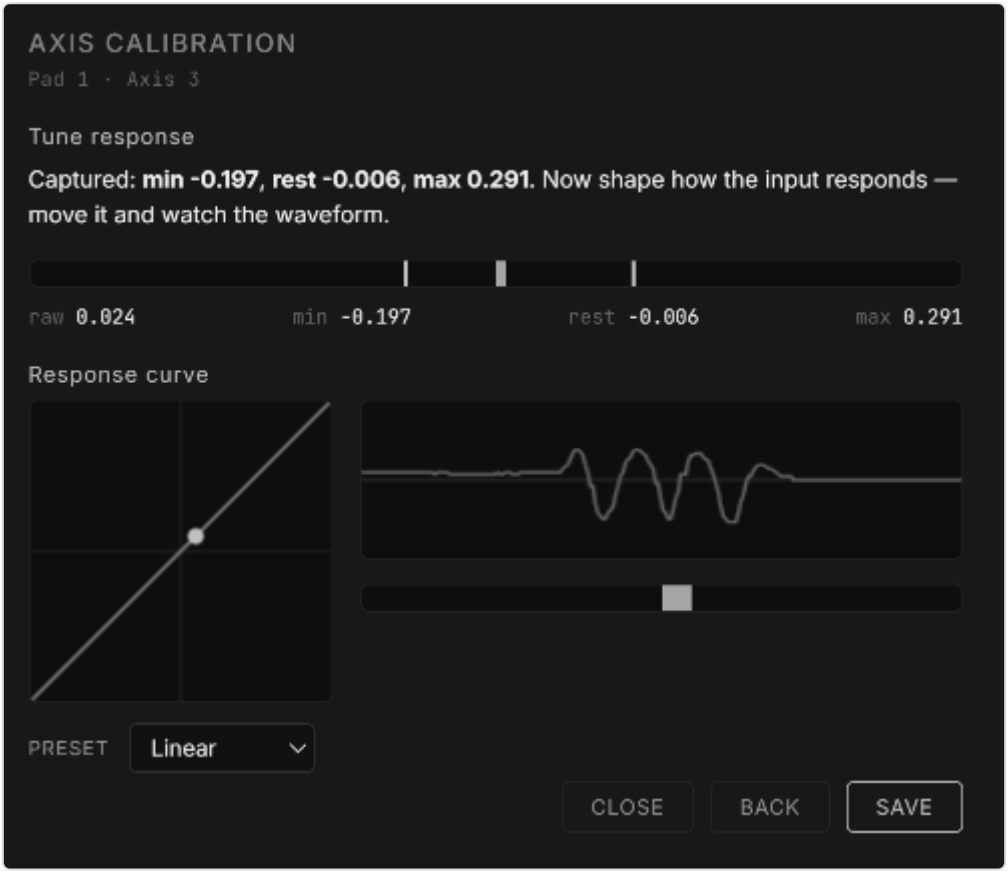
- **Show / Hide** — toggles the live view of every axis and button on connected controllers.
- **Null Drift (5s)** — samples all axes for 5 seconds and stores the average as zero, nulling resting drift. **Clear Cal** removes it.



The per-axis **"C"** button opens the calibration wizard — it reads **C·ƒ** when a live filter chain is running on that axis, and turns cyan once the axis is calibrated. The same wizard is available on every OSC / MIDI / FreeD input from the Other Inputs panel (section 12). Steps 1–3 capture the axis **max**, **min**, and **rest** so uneven hardware maps to a clean −1…+1. On the live bar the **blue** marker is the captured max and the **pink** marker is the captured min.

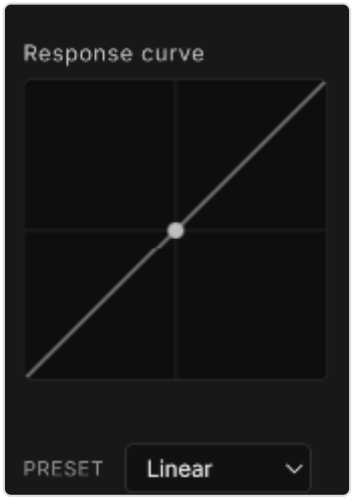


The **Tune response** step shapes the feel, with a live waveform and curve preview. **Skip** jumps straight to the tuner; **Redo step** re-captures the current step.

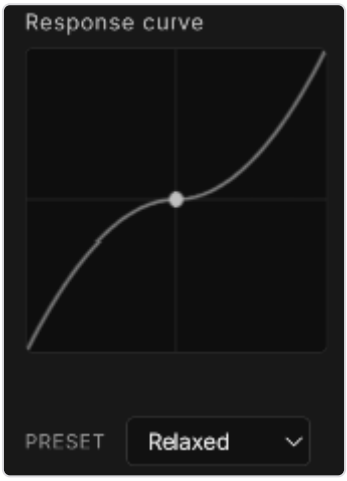


Tune response

Pick a **preset** — **Linear** (a straight 1:1 response)…



…or **Relaxed** / **Aggressive**, or drag the curve to a custom shape. Gentle-at-centre curves give fine control near rest; steeper ends give quicker throws.



Signal Filter Lab

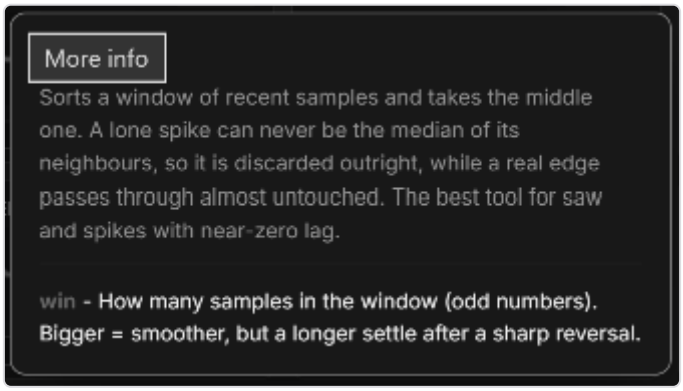
Rough or noisy inputs — like a load-cell Grip — can be cleaned up here, live, without touching the device firmware. **Capture** records a few seconds of the axis: the **pink** line is the raw signal, the **cyan** line is the filtered result.



+ Signal filter opens a 3×3 grid of the nine filters — Median, Moving average, EMA low-pass, Double-EMA, Kalman, 1-Euro, Hampel, Deadband and Slew limit — each previewing its effect on your captured signal, with editable parameters. **+ Add** appends a filter to the live chain, so several can stack in order and the combined result drives UE in real time.



The ⓘ icon on each filter explains what it does and what every parameter changes, so you can tune by feel rather than guesswork.



Save stores the current chain as a named **filter set** you can **Load** onto any axis or share between machines. **Scan 10s** records raw + filtered + output to a JSON log with a noise-reduction figure, for before/after comparisons.

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12. Other inputs — OSC, MIDI, FreeD



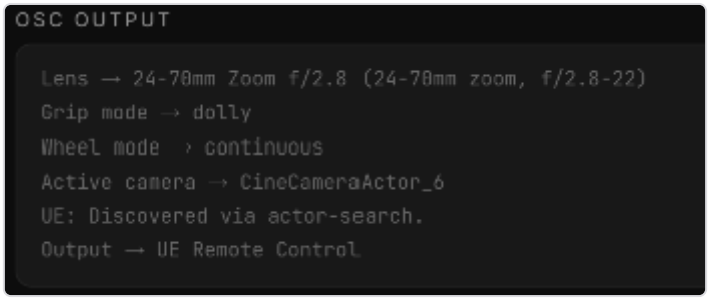
- **OSC in** — accept OSC from external devices (e.g. the Rexy Wheels). Set **port**, an optional **prefix** filter, and the input **min/max** range. **Sample 10s** records every OSC message for ten seconds and exports a JSON log — the ground-truth capture for diagnosing a device.
- **MIDI** — **Enable MIDI** reads MIDI controls (knobs, faders, notes, pitch-bend) into the same input path as gamepads, so they bind the same way.
- **FreeD in** — accept an incoming FreeD tracking stream.

All three feed the same binding system: enable the source, then **Bind** any control to it.

Calibrate and filter any input. Each live input row has a **C** button that opens the same calibration wizard and **Signal Filter Lab** as the gamepad axes (section 11) — so an OSC, MIDI or FreeD stream gets the full treatment: capture its range and rest, shape a response curve, and stack a live filter chain to clean up a noisy feed. The button reads **C·ƒ** when a filter chain is running and turns cyan once the input is calibrated.



13. OSC output log



A live feed of what the bridge is sending and status messages — handy for confirming a control is reaching UE and for spotting warnings.

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14. Remote viewing — phone, tablet, or a second laptop

The Rexy Bridge window you see on the PC is not the bridge itself — it's just a web control panel talking to the bridge over the network. That means **any device on the same Wi-Fi/LAN can open the exact same panel in a browser** and drive the same cameras, no install required. Think of it as a wireless director's monitor / second operator station.

How to open it on another device

1. On the PC running Rexy Bridge, find its local IP address. Open a Command Prompt and type `ipconfig` — use the **IPv4 Address** (something like `192.168.1.42`).
2. On the phone/tablet/laptop — connected to the **same network** — open a browser and go to:

`http://192.168.1.42:9000` *(swap in your PC's IP)*

1. The full control panel loads. It's live and **in sync**: a mode change, a slider, or a binding made on one device shows on all of them, because every panel is a client of the one bridge.

Good to know

- The bridge serves on **port 9000** to the whole LAN, so no extra setup is needed — just the IP and that port.
- Physical **wheels / gamepads / MIDI** are read by whichever device they're plugged into. A tablet with no controller is perfect as a **touch surface** (sliders, mode buttons, focus, takes) or a monitoring view, while the PC handles the hardware wheels.
- If the page won't load: both devices must be on the **same network** (not guest Wi-Fi / not a VPN), and a PC firewall prompt for Rexy Bridge must be **allowed** the first time.
- It's the same panel, so everything in this guide applies identically on the remote device.

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15. Rendering your move out of Unreal

The editor viewport is not what your final render looks like. The viewport runs in real time at a variable frame rate with little or no motion blur, so camera moves can look steppy or strobe

against high-contrast detail — bright lights, foreground objects. A proper render is done offline, one frame at a time, and looks considerably smoother. Judge your move from a render, not the viewport.

Capturing the move

Rexy Bridge drives the camera live, so first bake that motion into a Level Sequence:

1. **Window → Cinematics → Take Recorder**, add your Cine Camera Actor (and crane) as sources.
2. Press **Record**, perform or play back the move, then stop.

Take Recorder produces a **master sequence** plus a `_Subscenes` folder holding one sequence per recorded actor:

```
Scene_1_17                        <- the master: render THIS
Scene_1_17_Subscenes/
    CineCameraActor2_Scene_1_17
    CameraRig_Crane_Scene_1_17
```

> ⚠ Render the **master**, not a subscene. A subscene holds one actor's animation with no camera and no Camera Cuts track — the render will report **"no shot"** and render from the wrong viewpoint.

Open the master in Sequencer and confirm it has a **Camera Cuts** track pointing at your Cine Camera Actor (scrubbing should show the camera's view). If it's missing: **+ Track → Camera Cut Track**, then **+ Camera** → your camera, and save.

Rendering — the setting that matters

Window → Cinematics → Movie Render Queue → + Render, pick the master sequence, then click the config link:

Setting	Suggested	Why
Temporal Sample Count	**8**	**The important one.** Accumulates 8 samples per output frame to produce real motion blur. This is what smooths small steps in camera motion — at 1 you get none.
Spatial Sample Count	1	Anti-aliasing within a frame; doesn't affect motion.
Output Resolution	1280×720 for tests	Keeps memory use and render time sane.
Output Type	JPG Sequence	Small and dependency-free. EXR is huge; MP4/ProRes need an encoder configured.
Custom Start/ End Frame	~120 frames	4 seconds is plenty to judge smoothness.

Raise resolution and sample count for a final render; keep them low while testing.

If a render crashes, it's almost always memory — MRQ holds every temporal sample of a frame at full resolution at once. Drop the resolution first, then Temporal Samples to 4, and close other applications (including any second Unreal instance).

If Take Recorder stops with "Timecode rolled over"

Unreal generates timecode from your PC's clock by default. Those wall-clock values become very large frame numbers, which can overflow while a take is being finalised — reported as a rollover.

Fix: *Project Settings* → *Engine* → *General Settings* → *Timecode* → untick **Generate Default Timecode**, leave **Timecode Provider** empty, and restart the editor. Unless you're syncing to external gear, you don't need engine timecode at all.

A note on input smoothness

Motion blur hides a lot, but it can't invent detail that was never sampled. If a move still looks stepped in a render, check the input itself — the **Signal Filter Lab** (section 11) shows the raw device signal, and the tune-scan export records its true resolution and update rate. A low-resolution or slow-reporting control will quantise the move before Rexy Bridge ever sees it, and no amount of filtering fully recovers it.

Questions or bug reports: rob@realprogear.com